Chi-Sheng (Michael) Chen

Physiological Time-Series ML • Biosignal Foundation Models • LLM Agents • Clinical AI

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Research Affiliate at Harvard Medical School & BIDMC building interpretable ML for clinical biosignals (EEG, EMS trauma triage) and biosignal foundation models; EEG depression-treatment model in routine clinical use at Taipei VGH. Prior: CTO of Neuro Industry, Digital IC Engineer at MediaTek. Reviewer for NeurIPS/ICML/AAAI/KDD and 33 journals.

Education

National Taiwan University (NTU) — Taipei, Taiwan
M.S., Computer Science & Bioinformatics (GPA 3.74/4.30) — Adv. Prof. Cheng-Ta Li, MD & Prof. Chung-Ping Chen — *Sep 2019 – Jun 2021*

National Chiao Tung University (NCTU) — Hsinchu, Taiwan
B.Eng. in Electrophysics & B.S. in Interdisciplinary Science (dual degree) — *Sep 2015 – Jun 2019*

Selected Experience

Harvard Medical School & Beth Israel Deaconess Medical Center — MA, USA
Research Affiliate — Advisor: Prof. Gabriel Brat — *Nov 2024 – Now*

- Real-time EMS trauma-triage **multi-agent** pipeline: ASR (Whisper) → LLM-agent event extraction (RAG, tool use, task planning) → structured prediction.

Neuro Industry, Inc. — CA, USA
Co-Founder & CTO — *Mar 2024 – Jan 2025*

- Built GCP MLOps pipeline training on 10K+ EEG samples; led 3 papers on quantum ML and EEG-to-image diffusion.

Omnis Labs — Remote
Co-Founder — *Sep 2024 – Now*

- AI-driven DeFi liquidity/risk curator across AMM and perp DEX (Hyperliquid, Curve, Steer, Aster); 3 research papers.

MediaTek Inc. — Hsinchu, Taiwan
Digital IC R&D Engineer — *Dec 2021 – Oct 2023*

- Microprocessor IP for flagship 5G SoCs; hardware-virtualization RTL design & UVM/SystemVerilog verification.

Agent & LLM Engineering

Berkeley RDI AgentX-AgentBeats — 3rd Place / 1,300+ teams (2026). *MadGAA*, a multi-agent OSCE medical evaluation system: role-specialized agents (examiner, patient, judge) + verifiable scoring rubric (RAG + tool use + verifier).

WhaleForce — US-stock LLM agent suite (deployed). Browser agent as an explicit PLAN→LOCATE→ACT→VERIFY state machine with fault-injection recovery; layered SEC 10-K extractor; 22 research agents + 11 placebo controls with lookahead-free backtests, per-provider cost ledger, and eval dashboards.

Robotics Workcell Optimizer (live). NL → spec → robot selection → 2D layout → 3D preview via multi-LLM abstraction (Claude/OpenAI/Gemini, validation-repair loop) + OR-Tools CP-SAT; ISO 13855 safety as inviolable hard constraints.

Also: *flyhypo* (connectome + PubMed + LLM → grounded, falsifiable neuron hypotheses); *paper-evidence* (verbatim-quote gate + cross-family judge to block LLM hallucination); two live NYCU RAG physics tutors (Next.js + Gemini + pgvector); OpenAI RLHF contract (SFT/DPO/reward modeling).

Stack: LangChain/LangGraph, ReAct & function calling, multi-agent orchestration, vLLM, LoRA/QLoRA, SFT/DPO/RLHF, RAG (pgvector/FAISS/Qdrant), OR-Tools, OpenAI/Anthropic/Gemini SDKs.

Selected Publications

(first/co-first author; full list on Google Scholar)

Chen, C.-S., et al. "Unraveling Quantum Environments: Transformer-Assisted Learning in Lindblad Dynamics." *Physical Review A* (2025).

Chen, C.-S., et al. "QEEGNet: Cross-Task & Cross-Dataset EEG Encoding with Quantum ML." *J. Signal Processing Systems* (2025); SiPS 2024.

Chen, C.-S., et al. "Quantum Adaptive Self-Attention for Financial Rebalancing on AMMs in DeFi." *IEEE ICASSP* 2026 (oral).

Chen, C.-S., et al. "Large Cognition Model: Towards a Pretrained EEG Foundation Model." *arXiv:2502.17464* (2025).

Li, C.-T., **Chen, C.-S.**, et al. "Prediction of Antidepressant Responses to Non-Invasive Brain Stimulation from Frontal EEG." *J. Affective Disorders* (2023, IF 6.5).

Professional Service

Conference Reviewer: NeurIPS 2026, ICML 2026 (Gold Reviewer), AAAI 2027, KDD 2026, MICCAI 2026, ICASSP 2026, NLDL 2027.

Journal Reviewer (33 journals): incl. TPAMI, TNNLS, IEEE JBHI, National Science Review (OUP), npj Quantum Information, Scientific Reports, Elsevier Neural Networks. **PC:** QNRL @ IEEE WCCI 2026.